Data Visualization Project – Milestone 1

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I. Introduction

This document presents the first milestone of our data visualization project. The goal of this milestone is to define the topic of the visualization, evaluate the dataset that will be used, and perform preliminary exploratory analysis in order to assess the feasibility of the project. We introduce the dataset, then explain the problematic that motivates the visualization, present the first insights obtained from exploratory data analysis, and discuss related work and sources of inspiration.

II. Dataset

The dataset "Geospatial Environmental and Socioeconomic Data" [1] from Kaggle provides 12 datasets covering energy production, land use, and environmental indicators. The datasets have global scope, originate from various sources, and are provided in different formats. Our project uses 3 of the 12 datasets, described below.

a) Power Plants: Being the fourth dataset published by the World Resources Institute (WRI) and contains information about power plants worldwide. It contains 29,910 rows and 24 columns, including the plant name, country, coordinates, capacity in megawatts, and energy type. Additional columns exist but will not be used due to high rates of missing values. The data quality is generally good: it is a well-maintained, regularly updated global registry. Since the dataset is provided in CSV format and our columns of interest contain no missing values, no preprocessing is required.

b) Deforestation: Then choosing the eighth dataset that is provided by the Global Forest Watch (GFW) and covers deforestation worldwide from 2001 to 2019. It is a geospatial raster image in TIF format where each pixel represents a $10\,\mathrm{km}^2$ area, labeled with one of five deforestation drivers:

1) Commodity-driven deforestation
2) Shifting agriculture
3) Forestry
4) Wildfire
5) Urbanization

When the value is 0, it means that no deforestation was detected and if the value is below 0, it means no data. The dataset is complete with no missing values. However, since it is a raster image, preprocessing would be required: we will convert it to a CSV file such that each row represents a pixel, with columns for coordinates and the deforestation label. We also filter data with value and we add country information. Then, for visualization purposes, it can also be relevant to convert to GeoJSON that allow for ease of use with D3.

c) Population: Finally we took the fifth dataset, which provides gridded population estimates at a $250\,m^2$ resolution, published jointly by the European Commission Joint Research Centre (EC JRC) and the Columbia University Center for International Earth Science Information Network (CIESIN). It represents global population density for the year 2015 and is provided as a TIF raster image, where each pixel encodes the population density of its corresponding area. As with the deforestation dataset, preprocessing is required: we will convert it to a CSV file in which each row represents a pixel, with columns for coordinates and population density A conversion to GeoJSON can also be made here to use with D3. The difference is the downsampling to a $10\,\mathrm{km}^2$ resolution, which is the same as the deforestation dataset, to allow for easier merging and visualization.

III. Problematic

Deforestation is a major environmental challenge closely linked to energy production, land use, and regional socioeconomic factors. This visualization aims to explore the relationship between the distribution of power plant energy types across regions and the potential risk of increasing deforestation.

By combining geospatial, environmental, and socioeconomic indicators from the dataset, the objective is to identify regions where the current energy infrastructure may be associated with higher deforestation pressure. The visualization will then focus on how different types of power plants (such as fossil fuel, hydroelectric, or renewable sources) are distributed geographically and how these patterns could correlate with environmental indicators related to deforestation. Certain type of energy infrastructures can indirectly contribute to deforestation through land clearing, infrastructure development, resource extraction, or increased human activity.

Thus, by analyzing these patterns, the visualization seeks to highlight which regions may be more vulnerable to future forest degradation. It then brings us to the motivation for studying this topic, which would be the potential growing need to balance energy production with environmental sustainability. Understanding how energy infrastructure relates to deforestation risk can help policymakers, researchers, and environmental organizations better anticipate the environmental

impact of energy planning. Identifying regions where specific energy profiles coincide with deforestation indicators may support more informed and sustainable energy strategies. The target audience would include policymakers, environmental researchers, data scientists, and stakeholders involved in energy and land management. More broadly, it is also intended for individuals interested in understanding the environmental consequences of energy development. And finally, the key insight the audience should gain is an understanding of how the regional composition of energy production may relate to the deforestation risk, allowing them to identify geographic areas where energy infrastructure and environmental pressures intersect. This perspective can contribute to more sustainable planning and decision-making regarding both energy systems and forest conservation.

IV. Exploratory Data Analysis

As describing in the dataset section, the preprocessing was mostly about transforming TIF images to CSV file.

The first interesting graph in Figure 1 is for the top-eight country power plant counts, showing the distribution of the primary fuel used in the power plants. It's interesting to see that the countries listed are located in different parts of the world and face different constraints. For example, we see that Canada, which has a lot of big rivers, has more power plants using water than others countries.

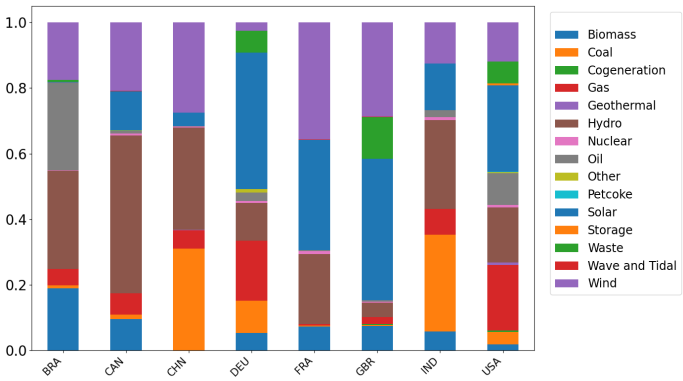

Fig. 1: Country distribution of the different types of power plants

Figures 2 and 3 yield an interesting observation: although the three primary power plants by count are respectively hydro, wind and solar, which account cumulatively for 61% of power plants, they only produce 25% of global energy. On the contrary, only 15% of power plants are Coal or Oil but they produce 62% of energy worldwide. This uneven distribution if energy density by plant can lead to interesting observations and visualizations when coupled to geographic and demographic data.

The array I show the summary statistics of annual electricity generation (in GWh) across power plants for selected years. The table reports the number of observations, central tendency (mean and median), dispersion (standard deviation), and distribution quartiles. The large gap between median and mean,

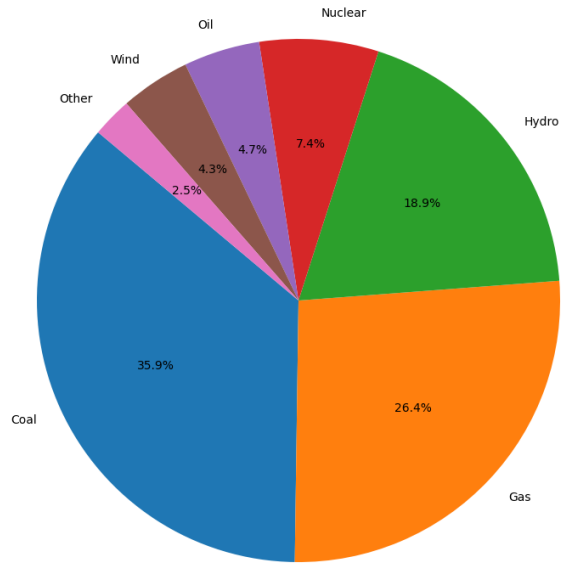

Fig. 2: Primary fuel distribution over the power produced by the power plants

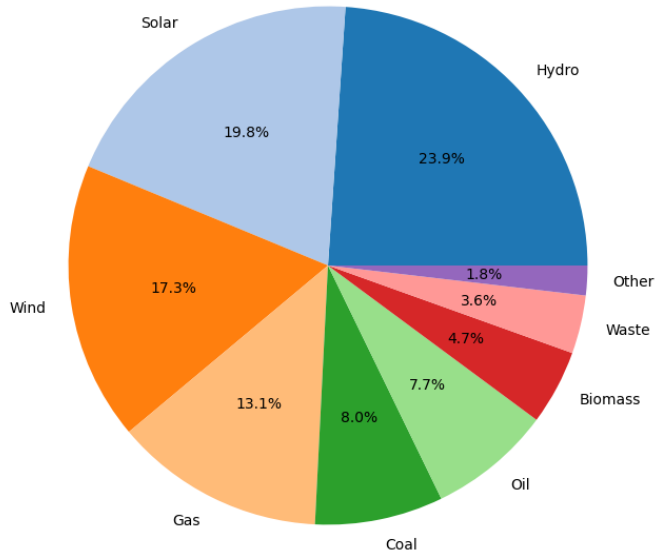

Fig. 3: Primary fuel distribution over the number of power plants

along with the high standard deviation and extreme maximum values, indicates a highly skewed distribution where a small number of large power plants account for a disproportionate share of total generation. Negative minimum values suggest potential data inconsistencies or adjustments.

Metric	2013	2014
Count	6996	7440
Mean	689.63	664.84
Std	2330.87	2258.35
Min	-947.60	-989.62
25%	2.17	2.16
50%	27.03	23.01
75%	250.60	224.96
Max	50834.00	32320.92

TABLE I: Summary Statistics by Year

The table II show the summary statistics of installed power plant capacity (in MW). The substantial difference between the mean (186.29 MW) and the median (18.90 MW) indicates a highly skewed distribution, where most power plants are relatively small while a limited number of very large facilities significantly increase the average. The extremely high maximum value (22500 MW) further highlights the presence of large-scale power plants that dominate total installed capacity.

Statistic	Value
Mean Capacity (MW)	186.29
Median Capacity (MW)	18.90
Max Capacity (MW)	22500

TABLE II: Summary Statistics of Capacity

Another interesting fact is that 50% of the population lived on only 2% the areas in our dataset. Reminder that the dataset only contains a 10km^2 area over the world with a population of at least 1. The Figure 4 shows the distribution of the population in areas sorted by the highest lived area.

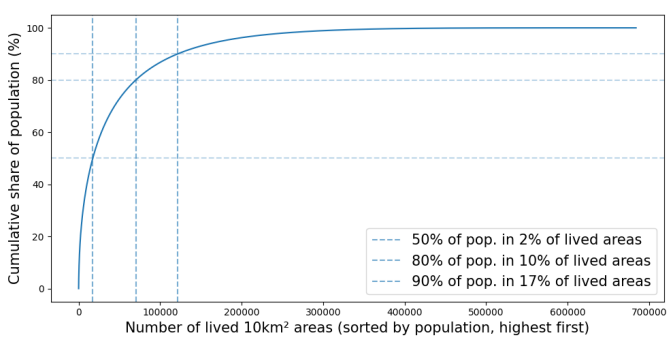

Fig. 4: Cumulative population on 10km^2 area sorted by most lived ones

V. Related Work

Our project is motivated and inspired by the planetary boundaries [2]. They are a framework that present indicators and limits on the safe operating space of human activity to achieve a sustainable and livable Earth. Planetary boundaries already offer a comprehensible visualization on climate and sustainability. This simple visualization consequently lacks extensiveness and real comprehension of the underlying indicators. Our work builds upon this framework by focusing on specific aspects of human impact. In particular, we consider data on power plants and deforestation, as they are in direct relation with radiative forcing and Human Appropriation of Net Primary Products (HANPP), which are two metrics of the planetary boundaries that are more than important but are often misunderstood and thus disregarded.

Other related work can be found on Kaggle, where multiple projects that use the same datasets as us are listed:

a) Ukraine [3]: This Kaggle notebook by Simon Lozada focuses geographically on Ukraine and uses data on urbanization, roads/railroads and power plants. This data is then superimposed on one map. This allows to visualize strategic positioning of power plants compared to cities and transport axes but also their vulnerability in the context of the recent invasion of Ukraine by Russia. While this work effectively highlights relationships between infrastructure elements, it remains focused on a single geographic region and does not explicitly relate these observations to broader sustainability metrics.

b) Power Plant Capacity [4]: This notebook by Jorge Garcia-Iniguez exploits data on power plants. It highlights on a map of Australia the location of every power plant and indicates their capacity by increasingly big circles on the map. This allows to view a sort of ”Energy production heatmap” that unsurprisingly coincides with the areas of high population density. Although this visualization clearly conveys the distribution of energy production capacity, it primarily emphasizes descriptive spatial patterns without linking them to environmental impact metrics. Our work extends this approach by integrating such data with indicators related to planetary boundaries, thereby providing additional context on the ecological implications of energy production.

References

[1] CatheTorres, “Geospatial environmental and socioeconomic data,” https://www.kaggle.com/datasets/cathetorres/geospatial-environmental-and-socioeconomic-data, 2021.
[2] Stockholm Resilience Centre, “Planetary boundaries,” 2023. [Online]. Available: https://www.stockholmresilience.org/research/planetary-boundaries.html
[3] S. Lozada, “Ukraine (kaggle notebook),” https://www.kaggle.com/code/simonlozada/ukraine, 2022.
[4] J. Garci-Iniguez, “Power plant capacity (kaggle notebook),” https://www.kaggle.com/code/jorgegarciainiguez/power-plant-capacity, 2022.